Project: Worked examples
Title: Column buckling (Perry–Robertson)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the compressive strength and resistance of a pin-ended strut by the Perry–Robertson formula, and its utilisation against an applied axial load. The Robertson constant comes from the strut curve.

WHAT IT CALCULATES	RELATION
Radius of gyration and slenderness	$r = \sqrt{(I/A)}$; $\lambda = L_E/r$
Euler buckling stress and load	$p_E = \pi^2 E/\lambda^2$; $F_E = p_E \cdot A$
Limiting slenderness, below which the strut is not reduced for buckling	$\lambda_0 = 0.2 \cdot \sqrt{(\pi^2 E/p_y)}$
Perry factor (Robertson constant a = 2.0, 3.5, 5.5, 8.0 for curves a–d)	$\eta = \max(0.001 \cdot a \cdot (\lambda - \lambda_0), 0)$
Compressive strength by the Perry–Robertson formula	$\varphi = (p_y + (\eta + 1) \cdot p_E)/2$; $p_c = p_E \cdot p_y/(\varphi + \sqrt{(\varphi^2 - p_E \cdot p_y)})$
Compression resistance and utilisation	$F_{c,Rd} = p_c \cdot A$; $u = F/F_{c,Rd}$

Basis: Compressive strength (Perry strut formula)[1](Annex C); Maximum slenderness[1](cl. 4.7.3.2); Allocation of strut curve[1](Table 23).

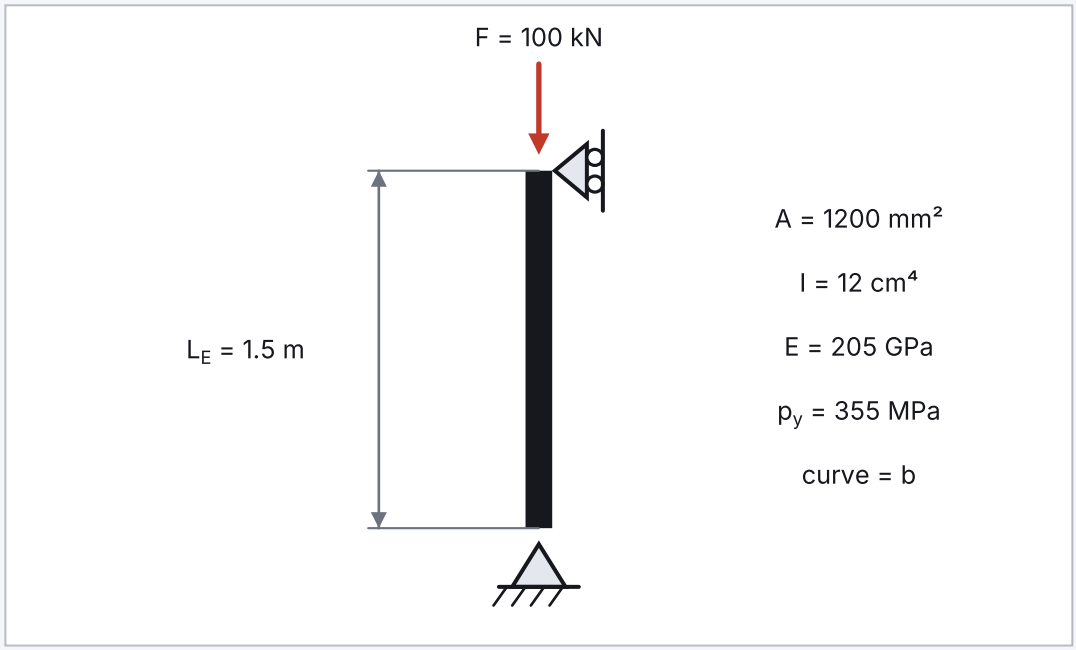


Figure 1 — A pin-ended strut under axial compression, effective length L_E. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
curve	Strut curve (selects the Robertson constant)	b		spec: BS 5950-1 Table 23: hot-finished tube, curve b	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_E	Effective length	1.5	m	drawing: DRG-ST-411 rev A	0.1 … 50 m
A	Cross-section area	1200	mm²	spec: SPEC-ST-02 section schedule	10 … 1000000 mm²
I	Second moment of area about the weaker axis	12	cm⁴	spec: SPEC-ST-02 section schedule	0.01 … 10000000 cm⁴
E	Young's modulus	205	GPa	spec: BS 5950-1 3.1.3: E = 205 GPa	50 … 450 GPa
p_y	Design strength (yield)	355	MPa	spec: SPEC-ST-02: S355	100 … 1000 MPa
F	Applied axial load	100	kN	assumption: Factored brace force	0 … 100000 kN

3. Assumptions

A1. The strut is pin-ended and prismatic, under axial load only. (Effective length carries the end restraint.)
A2. Buckling is about the axis of the given second moment. (The user gives the weaker axis.)
A3. Initial imperfection represented by the Perry factor $\eta = 0.001 \cdot a \cdot (\lambda - \lambda_0)$, with a from the strut curve. (Robertson's fit to strut tests.)
A4. F (Applied axial load) is taken as 100 kN. (Factored brace force)

4. Working

[8] $a_R = 3.5$ [1](Annex C)
= 3.5
Robertson constant for strut curve b
[9] r = sqrt(I / A)
= sqrt(12 cm⁴ / 1200 mm²)
= 10 mm
Radius of gyration
[10] $\lambda = L_E / r$
= 1.5 m / 10 mm
= 150
Slenderness
[11] p_E = pi^2 * E / λ^2 [1](Annex C)
= π^2 × 205 GPa / 150^2
= 89.923 MPa
Euler critical stress
[12] $F_E = p_E * A$
= 89.923 MPa × 1200 mm²
= 107.91 kN
Euler critical load
[13] λ_0 = 0.2 * sqrt(pi^2 * E / p_y) [1](Annex C)
= 0.2 × sqrt(π^2 × 205 GPa / 355 MPa)
= 15.099
Limiting slenderness
[14] η = max(0.001 * a_R * (λ - λ_0), 0) [1](Annex C)
= max(0.001 × 3.5 × (150 - 15.099), 0)
= 0.47215
Perry factor (initial imperfection)

[15] $\varphi = (p_y + (\eta + 1) * p_E) / 2$
$= (355\ \text{MPa} + (0.47215 + 1) \times 89.923\ \text{MPa}) / 2$
$= 243.69\ \text{MPa}$
Perry–Robertson φ

[16] $p_c = p_E * p_y / (\varphi + \text{sqrt}(\varphi^2 - p_E * p_y))$ [1](Annex C)
$= 89.923\ \text{MPa} \times 355\ \text{MPa} / (243.69\ \text{MPa} + \text{sqrt}((243.69\ \text{MPa})^2 - 89.923\ \text{MPa} \times 355\ \text{MPa}))$
$= 77.973\ \text{MPa}$
Compressive strength

[17] $F_{c,Rd} = p_c * A$
$= 77.973\ \text{MPa} \times 1200\ \text{mm}^2$
$= 93.567\ \text{kN}$
Compression resistance

[18] $u = F / F_{c,Rd}$
$= 100\ \text{kN} / 93.567\ \text{kN}$
$= 1.0687$
Utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{c,Rd}$	93.56 kN	Compression resistance
F_E	107.91 kN	Euler critical load
p_c	77.97 MPa	Compressive strength
λ	150	Slenderness
u	1.07	Utilisation
r	10 mm	Radius of gyration
p_E	89.92 MPa	Euler stress
λ_0	15.099	Limiting slenderness
η	0.47215	Perry factor

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Axial load against compression resistance	100 kN	93.56 kN	1.07	[1](Annex C)

7. Applicability

None.

8. Verification

Template struct.column-buckling@1.1.0 Content hash 9620a0aa904c2b4c Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Structural Use of Steelwork in Building—Part 1: Code of Practice for Design—Rolled and Welded Sections. BS 5950-1:2000. BSI; 2000.